The Poisson Distribution

ANSWER KEY

Conditions for a Poisson distribution

1 State the key conditions required for a random variable to be modeled by a Poisson distribution.

Events occur independently of one another; events occur at a constant average rate over the given interval (of time or space); two events cannot occur at exactly the same instant; the number of events in non-overlapping intervals is independent.

Computing Poisson probabilities

2 A call center receives an average of 4 calls per minute, modeled by a Poisson distribution with $\lambda = 4$. Find $P(X = 6)$, using $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$, to three decimal places.

$P(X = 6) = \frac{e^{-4}4^6}{6!} = \frac{(0.0183)(4096)}{720} \approx \frac{74.98}{720} \approx 0.104$.

3 Using the same call center ($\lambda = 4$ calls per minute), find the probability of receiving at most 2 calls in a given minute.

$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$. $P(X = 0) = e^{-4} \approx 0.0183$. $P(X = 1) = 4e^{-4} \approx 0.0733$. $P(X = 2) = \frac{16e^{-4}}{2} = 8e^{-4} \approx 0.1465$. Sum $\approx 0.0183 + 0.0733 + 0.1465 \approx 0.238$.

Mean and variance of a Poisson distribution

4 State the mean and variance of a Poisson distribution with parameter λ, and explain what is distinctive about this relationship compared to other common distributions.

For a Poisson distribution, both the mean and the variance equal λ: $E(X) = \lambda$ and $\text{Var}(X) = \lambda$. This is distinctive because for most other distributions, the mean and variance are independent parameters (e.g. the normal distribution has separate μ and σ^2) — for Poisson, a single parameter determines both.

Sums of independent Poisson variables

5 Two independent Poisson processes have rates $\lambda_1 = 3$ and $\lambda_2 = 5$ events per hour. If $X_1 \sim \text{Po}(3)$ and $X_2 \sim \text{Po}(5)$, state the distribution of $X_1 + X_2$, and find its mean and variance.

The sum of independent Poisson random variables is itself Poisson: $X_1 + X_2 \sim \text{Po}(3 + 5) = \text{Po}(8)$. Mean $= 8$; variance $= 8$ (since for Poisson, mean and variance are equal).

Poisson as an approximation to the binomial

6 Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the appropriate value of λ to use.

The Poisson approximation to the binomial is appropriate when n is large and p is small (a common guideline: $n > 50$ and $np < 5$), so that events are rare relative to the number of trials. In that case, X is approximately $\text{Po}(\lambda)$ with $\lambda = np$.

7 A factory produces 2000 items, each independently defective with probability 0.002. Using the Poisson approximation, find the probability that exactly 3 items are defective, to three decimal places.

$\lambda = np = 2000(0.002) = 4$. $P(X = 3) = \frac{e^{-4}4^3}{3!} = \frac{(0.0183)(64)}{6} \approx \frac{1.171}{6} \approx 0.195$.

Using cumulative Poisson probabilities

8 A shop receives customers at an average rate of $\lambda = 5$ per 10-minute interval. Find the probability of receiving more than 2 customers in a given 10-minute interval, given $P(X = 0) \approx 0.0067$, $P(X = 1) \approx 0.0337$, and $P(X = 2) \approx 0.0842$.

$P(X > 2) = 1 - P(X \leq 2) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)] = 1 - (0.0067 + 0.0337 + 0.0842) = 1 - 0.1246$

Finding an unknown parameter from a given probability

9 A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.1353$. Find λ, to the nearest whole number, using $P(X = 0) = e^{-\lambda}$.

$e^{-\lambda} = 0.1353$. Taking natural logs: $-\lambda = \ln(0.1353) \approx -2.0$, so $\lambda \approx 2$.

Variance-based reasoning with the Poisson distribution

10 A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 3. Find λ, and then find $P(X = \lambda)$, to three decimal places.

Since variance $= \lambda$ and standard deviation $= \sqrt{\lambda} = 3$, we get $\lambda = 9$.
$P(X = 9) = \frac{e^{-9}9^9}{9!} = \frac{(0.0001234)(387420489)}{362880} \approx \frac{47790}{362880} \approx 0.132$.

Combining Poisson reasoning with real-world decisions

11 A website server crashes on average $\lambda = 0.5$ times per week. The IT team wants to know the probability of at least one crash in a given week, given $P(X = 0) \approx 0.6065$. Should the team be concerned if their target is to keep this probability below 30%?

$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.6065 = 0.3935$, or about 39.35%. Since this exceeds their target threshold of 30%, the team should be concerned — there is roughly a 39% chance of at least one crash in a given week, above their comfort threshold, suggesting they should investigate ways to reduce the average crash rate λ.

Combining a Poisson approximation with conditional probability

12 Machine faults occur at an average rate of $\lambda = 1.5$ per day. Find the probability that exactly one fault occurs today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.2231$ and $P(X = 1) \approx 0.3347$.

$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.2231 = 0.7769$. Conditional probability: $P(X = 1 \mid X \geq 1) = \dfrac{P(X = 1)}{P(X \geq 1)} = \dfrac{0.3347}{0.7769} \approx 0.431$.

Combining Poisson processes over different time intervals

13 A radioactive source emits particles at a rate of $\lambda = 2$ per second, modeled by a Poisson distribution. Find $P(X = 3)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.

For a 3-second interval, the parameter becomes $\lambda = 2 \times 3 = 6$. $P(X = 3) = \dfrac{e^{-6}6^3}{3!} = \dfrac{(0.002479)(216)}{6} \approx \dfrac{0.5354}{6} \approx 0.089$.

14 Explain why the Poisson distribution is not an appropriate model for the number of heads obtained when flipping a fair coin 10 times, referencing the conditions required for a Poisson model.

The number of heads in 10 coin flips is naturally modeled by a binomial distribution $B(10, 0.5)$, not a Poisson distribution, because a Poisson model requires events to be rare (a small probability of occurrence per trial, with np moderate while n is large) — here $p = 0.5$ is far too large for the Poisson approximation to be valid, and there is a fixed, small number of trials ($n = 10$) rather than events occurring continuously over time or space.

Comparing two independent Poisson processes

15 Two independent Poisson processes have rates $\lambda_A = 3$ and $\lambda_B = 2$ per hour. Find the probability that, in a given hour, process A produces exactly 2 events and process B produces exactly 1 event, to three decimal places, given $P(X_A = 2) \approx 0.2240$ and $P(X_B = 1) \approx 0.2707$.

Since the two processes are independent, the joint probability is the product of the individual probabilities: $P(X_A = 2) \times P(X_B = 1) = 0.2240 \times 0.2707 \approx 0.0606$.